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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2026

Time: 03 Hours

Code: 100209 ***Subject: Basic Electronics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven questions only): **[2×7=14]**

(a) For a half-wave rectifier, if the peak secondary voltage is 20 V, the average DC output voltage (ideal diode, no filter) is approximately

(i) 6.37 V (iii) 12.74 V
(ii) 9 V (iv) 20 V

(b) An inverting amplifier has R1 = 10 kΩ and Rf = 100 kΩ. The gain is

(i) 10 (iii) −10
(ii) 11 (iv) −11

(c) The maximum rectification efficiency of an ideal full-wave rectifier is approximately

(i) 40.6% (iii) 81.2%
(ii) 57.6% (iv) 100%

(d) A capacitor filter connected across load in a rectifier tends to

(i) Increase ripple content (iii) Reduce ripple and smooth output
(ii) Decrease average output voltage (iv) Reverse the polarity

(e) A JFET has I_{DSS} = 10 mA, V_P = −5 V. If I_D = 3.6 mA, the approximate V_{GS} is

(i) −1 (iii) −2
(ii) −3 (iv) −4

(f) For a BJT, which biasing method generally provides better thermal stability?

(i) Fixed bias (iii) Collector-to-base bias
(ii) Voltage divider bias (iv) No bias

(g) Compared to BJT, a FET generally has

(i) Lower input impedance (iii) Zero output resistance
(ii) Higher input impedance (iv) No control terminal

(h) In an intrinsic semiconductor, the number of free electrons is

(i) Greater than the number of holes (iii) Equal to the number of holes
(ii) Less than the number of holes (iv) Zero

(i) Which one of the following is a direct band-gap semiconductor commonly used in LEDs?

(i) Silicon (iii) Gallium Arsenide
(ii) Germanium (iv) Carbon

(j) In an n-channel enhancement MOSFET, drain current starts increasing significantly when

(i) V_{DS} = 0 always (iii) V_{GS} exceeds threshold voltage
(ii) V_{GS} becomes more negative than threshold (iv) Gate current becomes very large

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Q.2 (a) Describe the enhancement-type MOSFET with a neat structure and explain why no channel exists at zero gate voltage. **[7]**

(b) What is an integrator circuit using an op-amp? Derive the output voltage equation and explain how it performs mathematical integration of the input signal. **[7]**

Q.3 (a) Convert the following numbers into: (i) $(1230)_4 = (....)_{10}$ (ii) $(12.0625)_{10} = (....)_2$ **[7]**

(b) For a JFET, I_{DSS} = 8 mA, V_P = −4 V. Determine the value of V_{GS} when the drain current is 2 mA. **[7]**

Q.4 (a) Calculate the subtraction of the following decimal numbers using the 1's and 2's complement method. (i) 5250 − 321 (ii) 20 − 1000 **[7]**

(b) Explain the difference between intrinsic and extrinsic semiconductors. Also compare n-type and p-type semiconductors with suitable energy band diagrams. **[7]**

Q.5 (a) Explain the breakdown mechanisms in a diode. **[7]**

(b) What is a clamper circuit? Explain the operation of a positive clamper and a negative clamper with waveforms. **[7]**

Q.6 (a) Explain the working and characteristics of a Zener diode. Why is it suitable for voltage regulation? **[7]**

(b) Explain the characteristics of a BJT in common collector configuration and compare it with CE and CB configurations. **[7]**

Q.7 (a) Explain the construction and working of a JFET. **[7]**

(b) Derive the expression for the gain of a non-inverting amplifier using ideal op-amp assumptions. **[7]**

Q.8 (a) Design a zero-crossing detector using an op-amp and draw its waveform. **[7]**

(b) Describe the operation of a center-tapped full-wave rectifier. Derive the expressions for output current, ripple factor, efficiency, and PIV. **[7]**

Q.9 Write short notes on any two of the following: **[7×2=14]**

(i) LED, photodiode and solar cell

(ii) Zener diode as a voltage regulator

(iii) Universal logic gates

(iv) What are drift current and diffusion current in a p-n junction

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